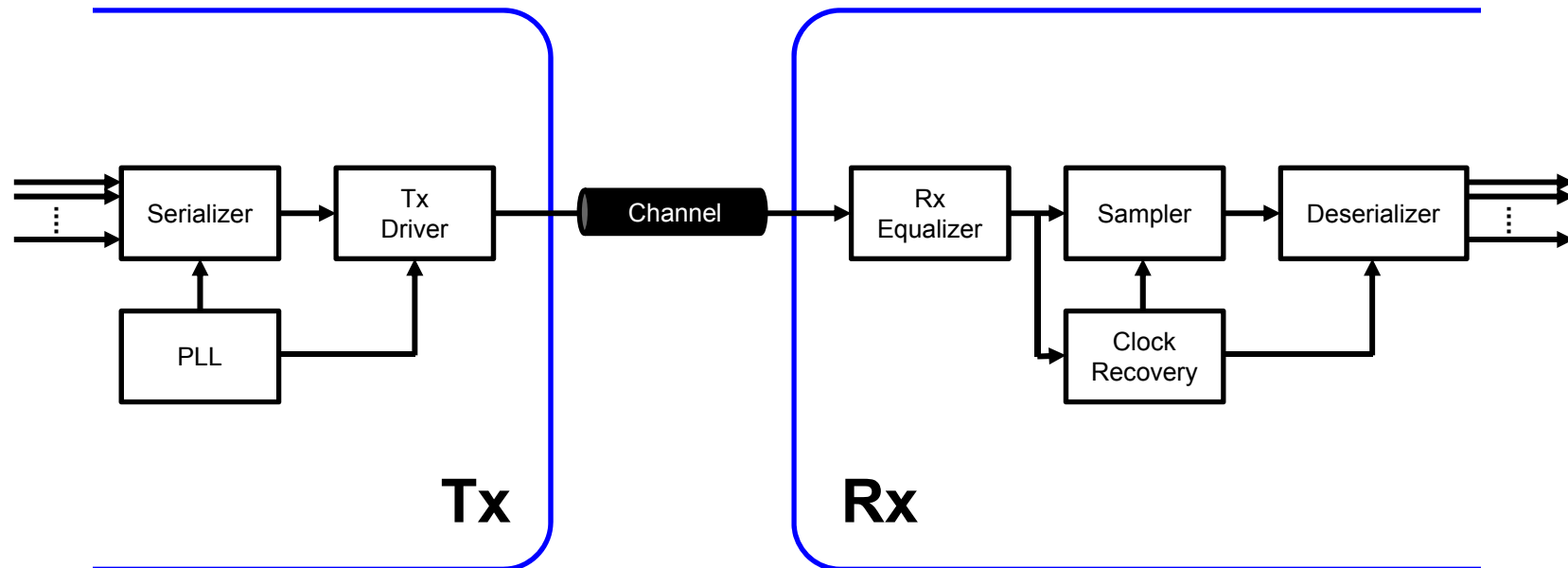


High-speed Serial Interface

Lect. 6 – TX Driver and Equalizer

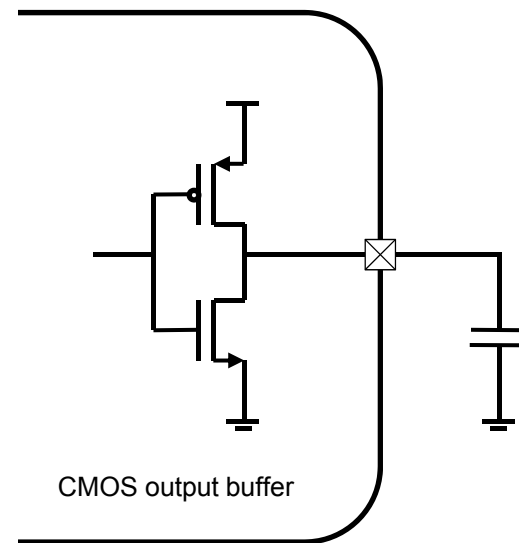
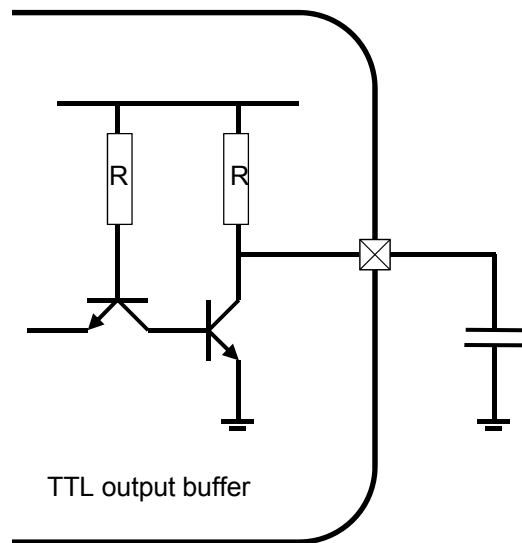
Block diagram

- Where are we today?



Classic output driver

- An inverter can be used as voltage-mode output driver



Classic output driver

- It is difficult to use inverter-style output driver in high-speed applications
 - Full-swing logic is speed-limited because of slow switching time of inverter-style driver
 - Impedance matching is not easy
 - Transistors have variable output resistances during output voltage transients

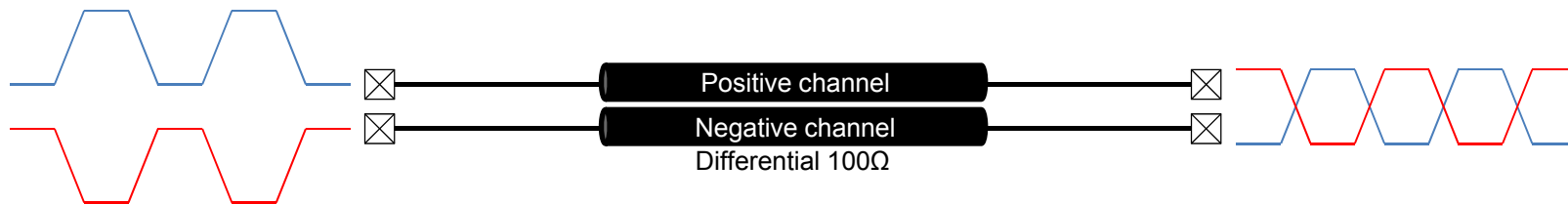
Single-ended signaling

- Signal is transferred via single channel
- Simple but ...
- Threshold should be generated in RX side.
 - Logic levels in TX may not be same as in RX side
 - Supply and ground levels are different for TX and RX sides
- Poor noise immunity
 - Noises are added while signals travel through channel



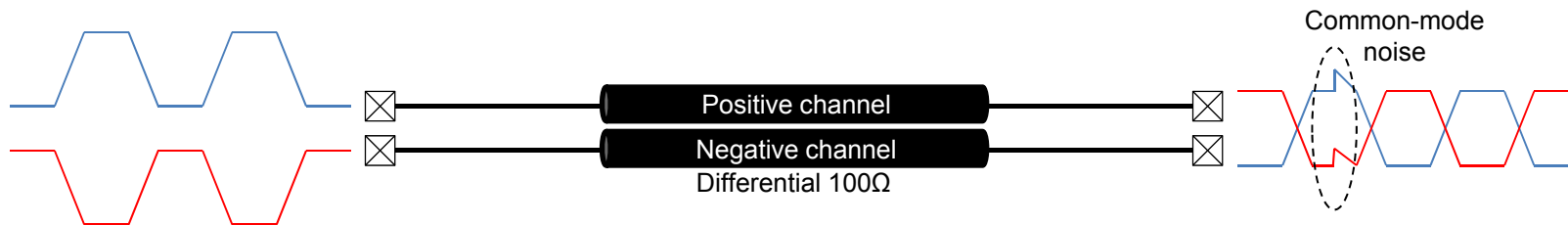
Differential signaling

- Differential signals are transferred via two adjacent channels
 - Each signal has opposite logic level
 - Ex) twisted pair, differential PCB lines



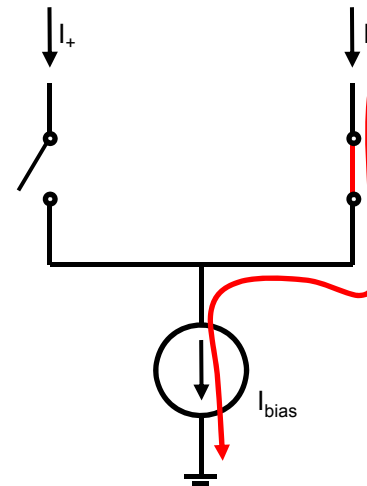
Differential signaling

- Larger signal swing and self-reference
 - $\text{Signal} = (\text{positive signal} - \text{negative signal})$
 - ➔ Decision margin enhanced
 - $\text{threshold} = (\text{positive signal} + \text{negative signal})/2$
- Common-mode noise rejection
 - Noise usually affects both positive and negative channels
 - Subtraction rejects common-mode noise



Current-mode driver

- Reduced switching time
 - Current-steering: Switching current path while source current is kept constant.
 - Switching time is reduced since current source is not turned-off
- Disadvantage
 - Differential signaling is required.
 - Static current causes static power consumption
 - Usually larger power consumption than voltage-mode

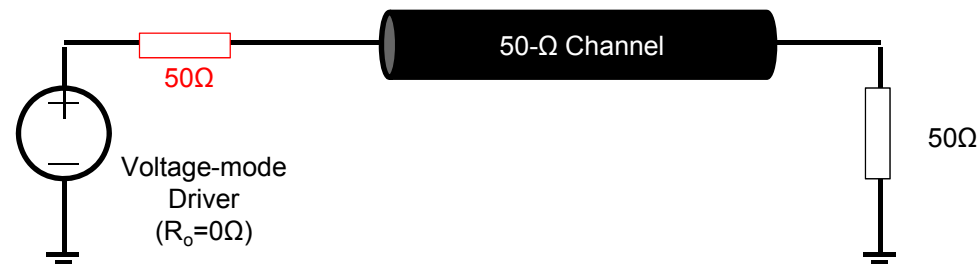


50- Ω termination

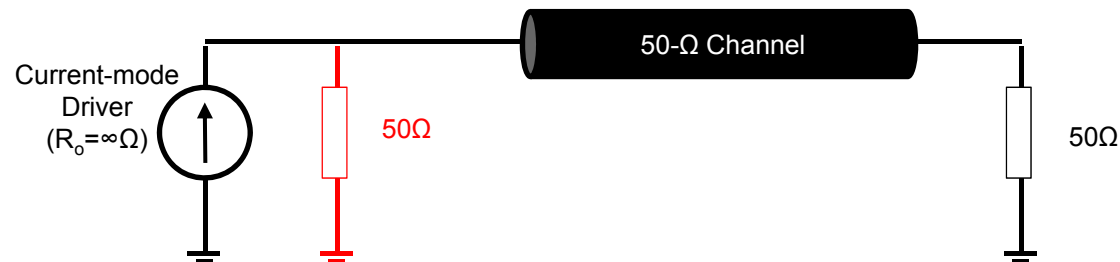
- Why 50 Ω ?
 - Historical issue
 - In early microwave systems, it was known that
 - 33 Ω shows best performance in power transfer
 - 75 Ω shows best performance in signaling
 - For convenience, 50 Ω was selected instead of medium value, 54 Ω
 - Nowadays, almost all high-speed instruments are 50 Ω -based
 - ➔ Significant for high-speed serial interface
 - In CATV systems, 75- Ω termination is still used

50- Ω termination

- Tx-side termination topology
 - Voltage-mode driver has small output impedance
 - Series termination

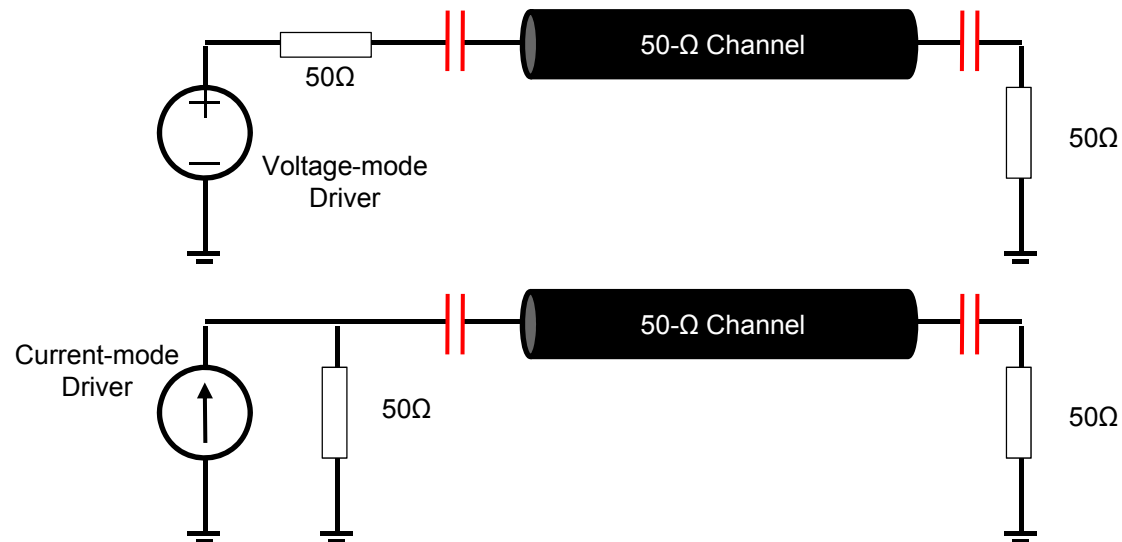


- Current-mode driver has large output impedance
 - Parallel termination



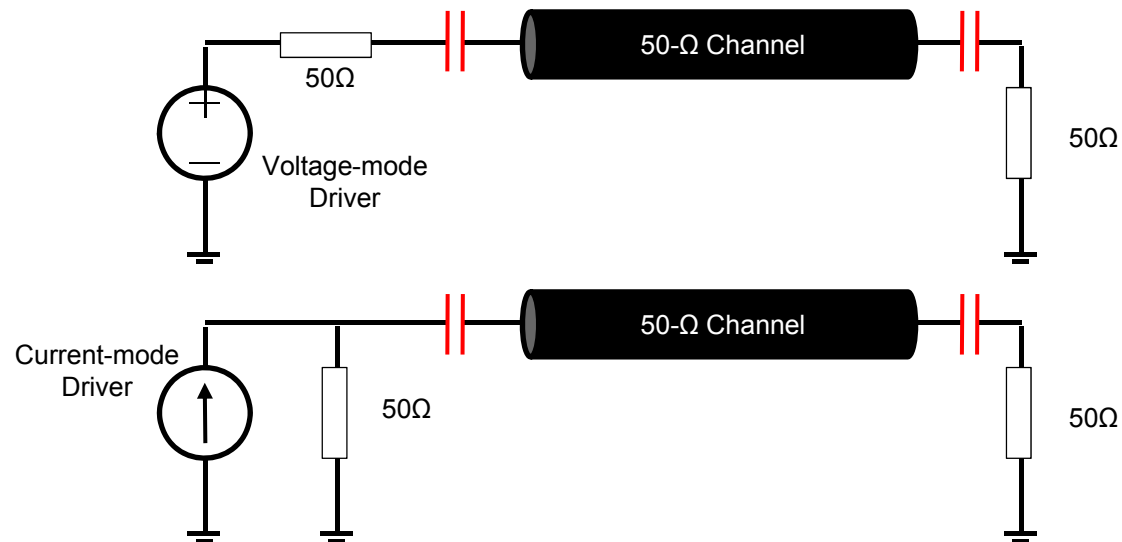
DC- and AC-coupling

- AC coupling with a series capacitor
 - Both TX and RX are possible
 - Common-mode voltage can be separately controlled in both side
 - Coupling capacitor can causes low-frequency loss
 - Capacitance $> 100\text{nF}$ is generally used.



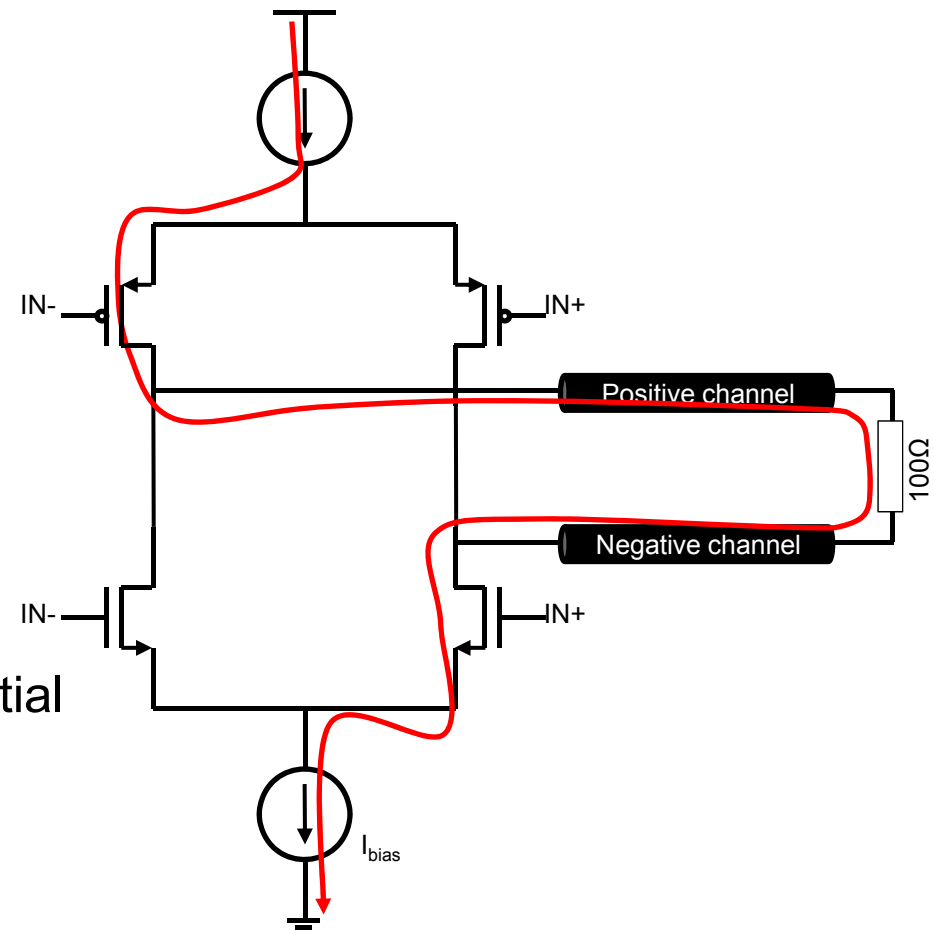
DC- and AC-coupling

- AC coupling cannot be used if consecutive identical bits are transmitted
→ 8B/10B coding for many standards



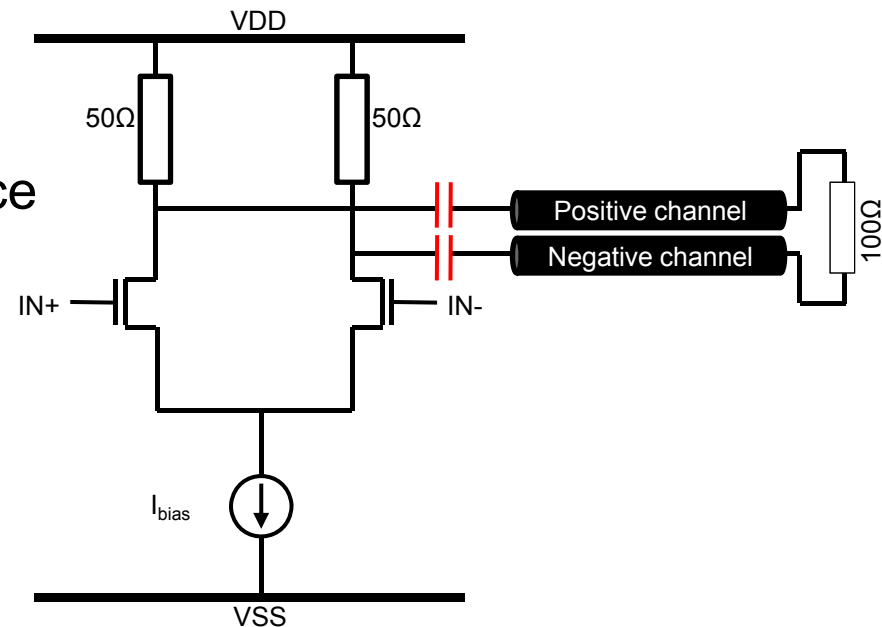
Push-pull driver

- 2 current sources
 - Current path switching
 - Upper and lower pairs
 - Same rising and falling time for each differential signal
 - Upper PMOS pair can be replaced by NMOS pair to enhance switching time
 - Head room problem in low-voltage technologies
 - Used in Low-Voltage Differential Signals (LVDS) standard
 - TX termination?



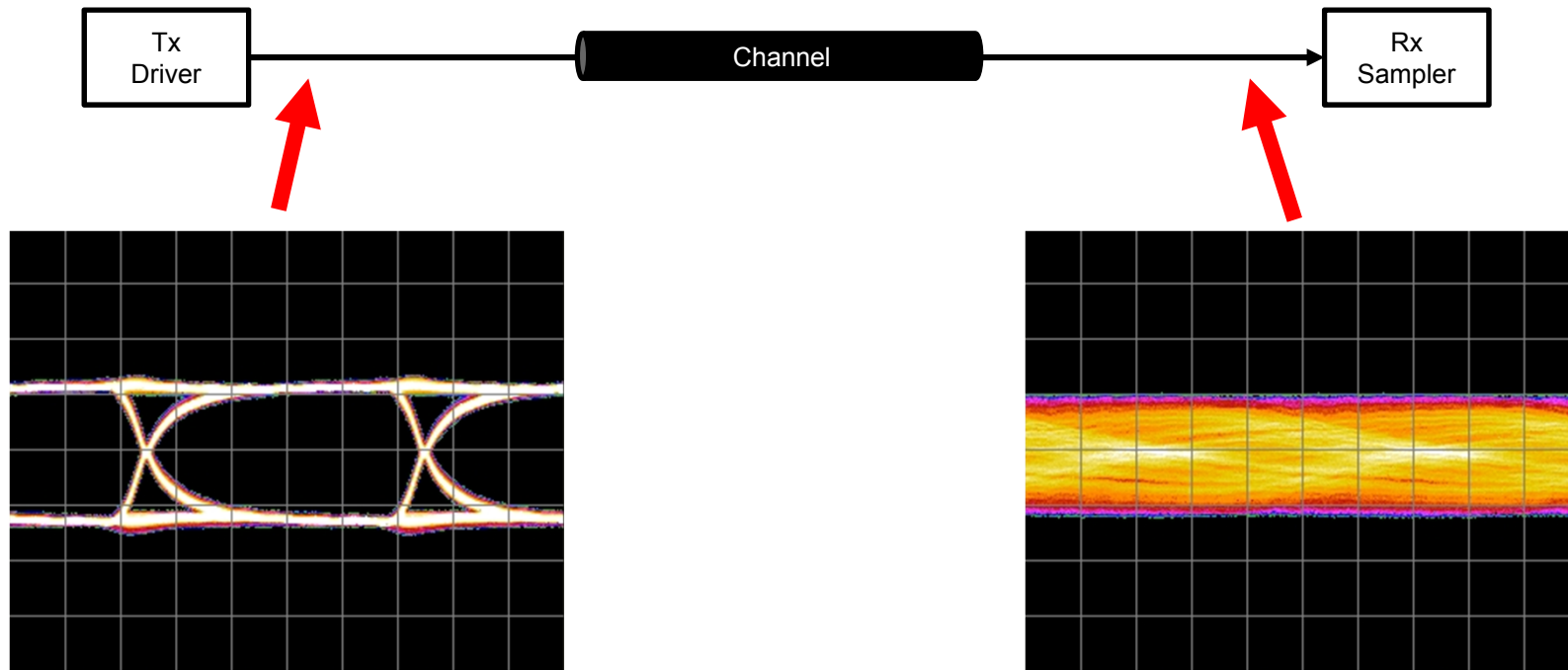
CML (Current-Mode Logic) driver

- Loaded by 50ohm resistor
 - Current steering
 - Both side are terminated by 50Ω
 - Output voltage can be both DC, AC-coupled
 - Used in most high-performance serial link



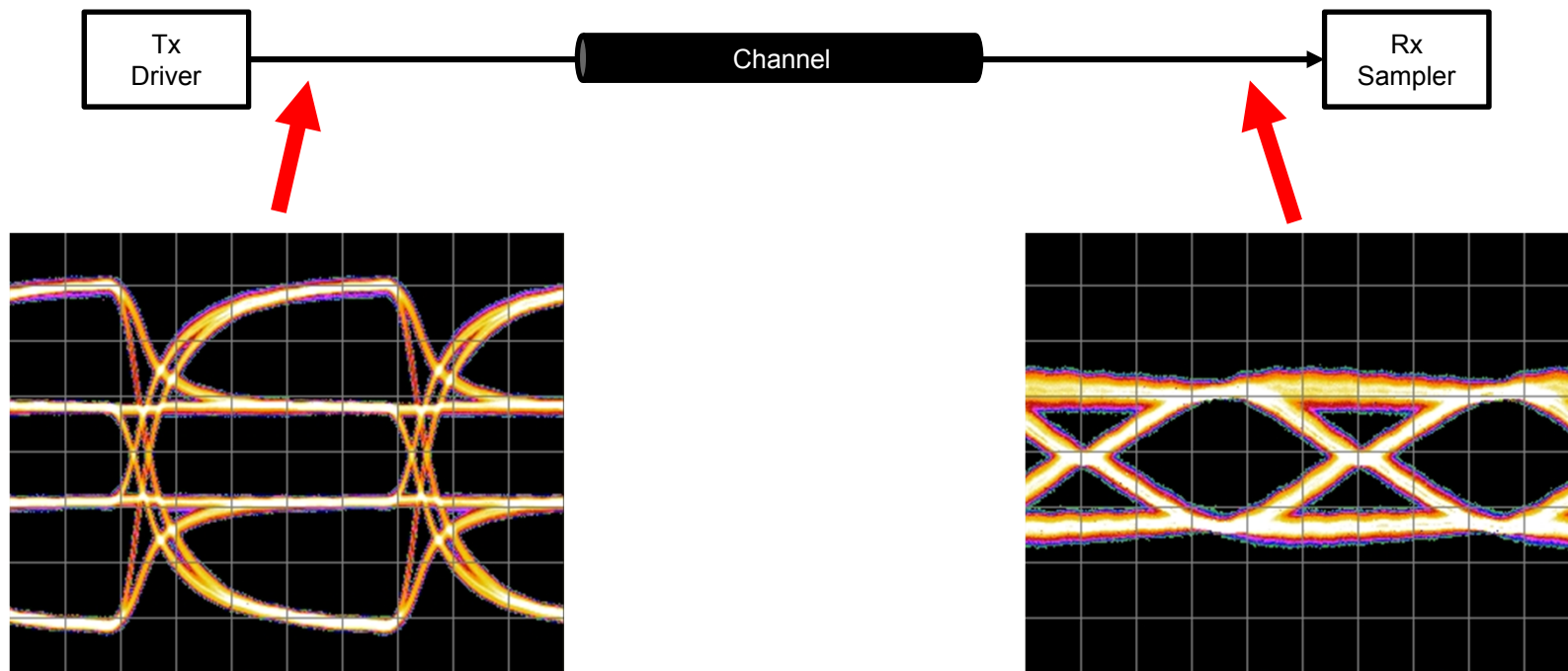
TX equalization

- Channel causes ISI on received signal.
 - High-frequency loss in channel $\rightarrow$ eye-diagram closed



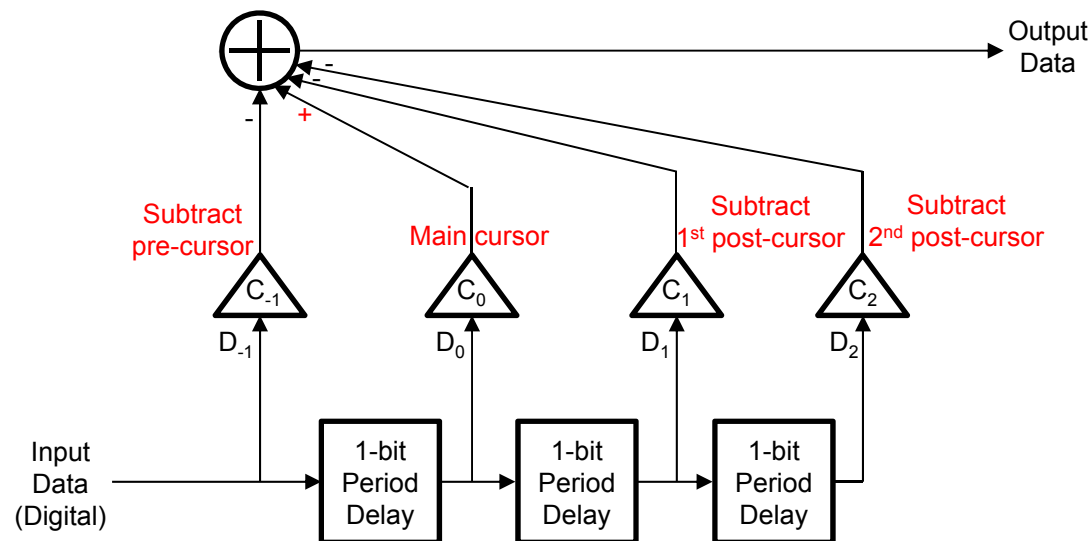
TX equalization

- TX driver can be also channel equalizer
 - TX driver can enhance high-frequency components before traveling through channel.



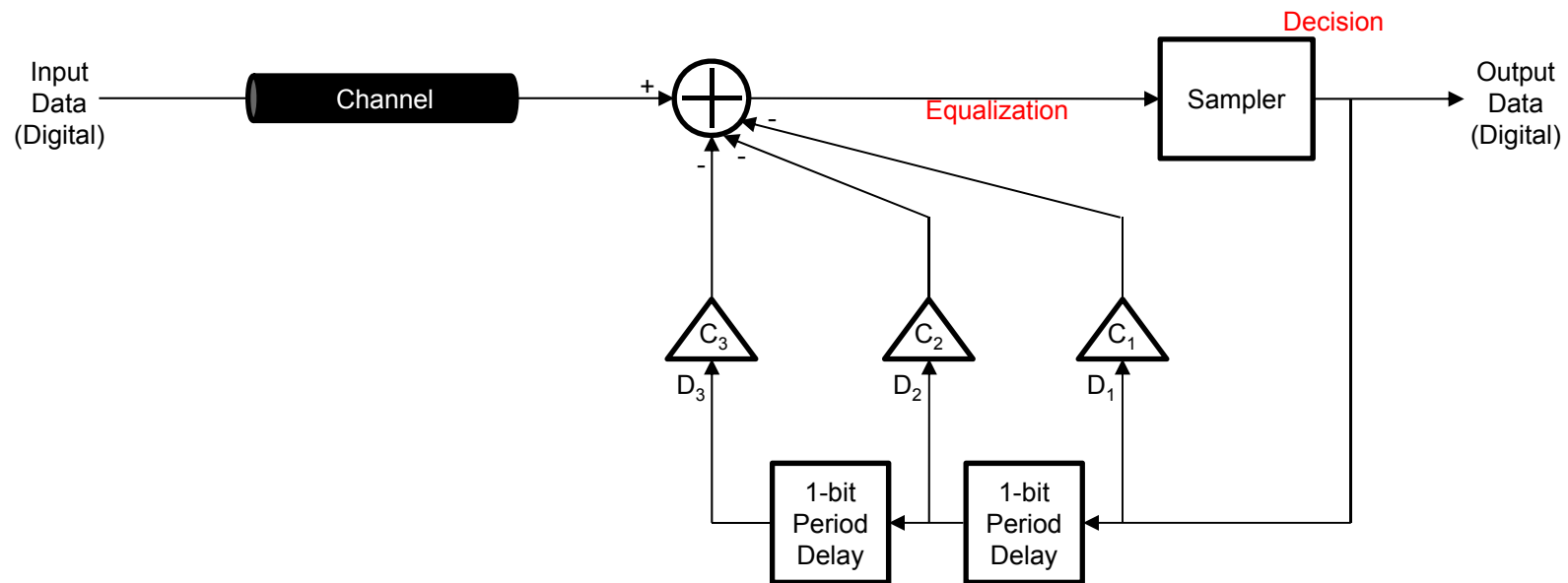
How to reject ISI?

- FIR filtering
 - Forcing cursors to 0 can be implemented by FIR filtering.
 - ISI can be removed since we know input data in TX-side
 - Tx-side FIR filtering can include pre-cursor



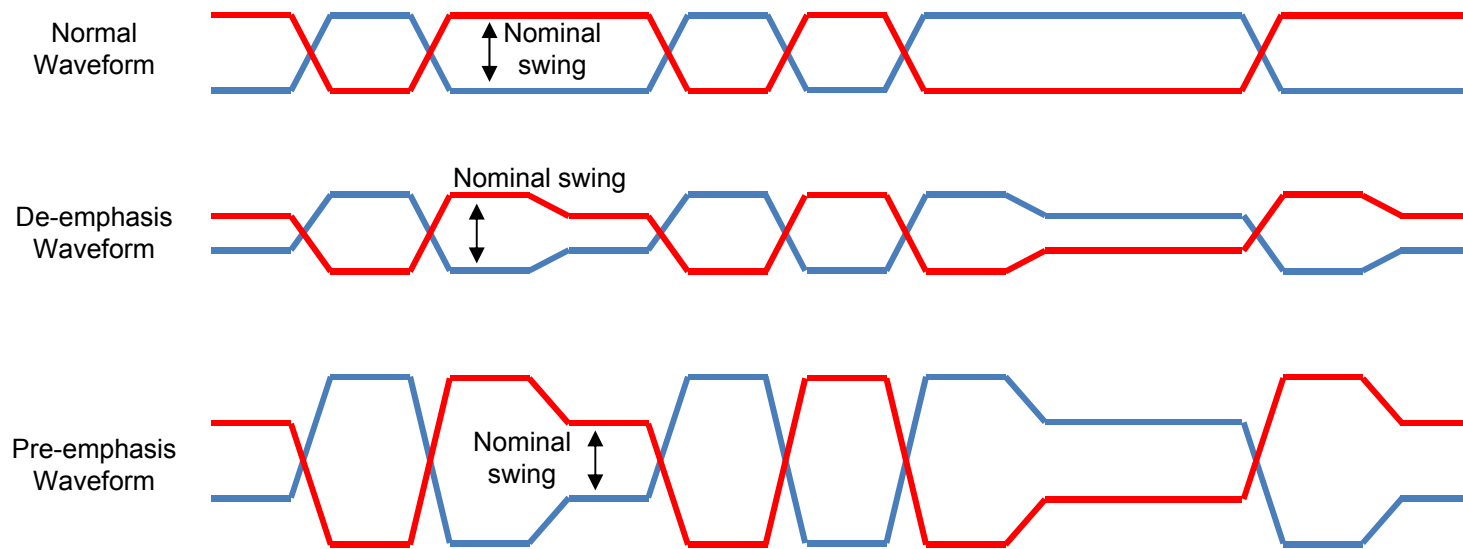
Feedforward vs Feedback

- DFE



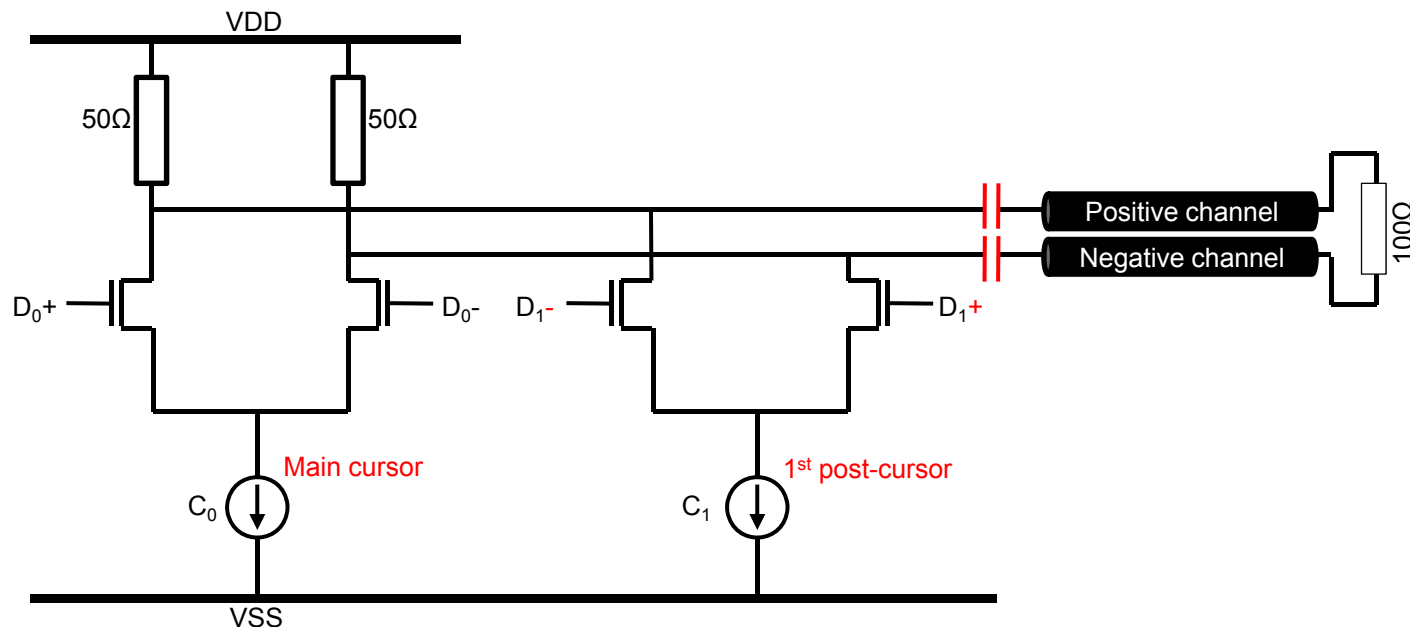
Pre-/De-Emphasis

- Tx FIR is often called Pre-/De-Emphasis
 - De-emphasis: to reduce low-frequency components
 - Pre-emphasis: to enhance high-frequency components
- High-frequency component is transition bits



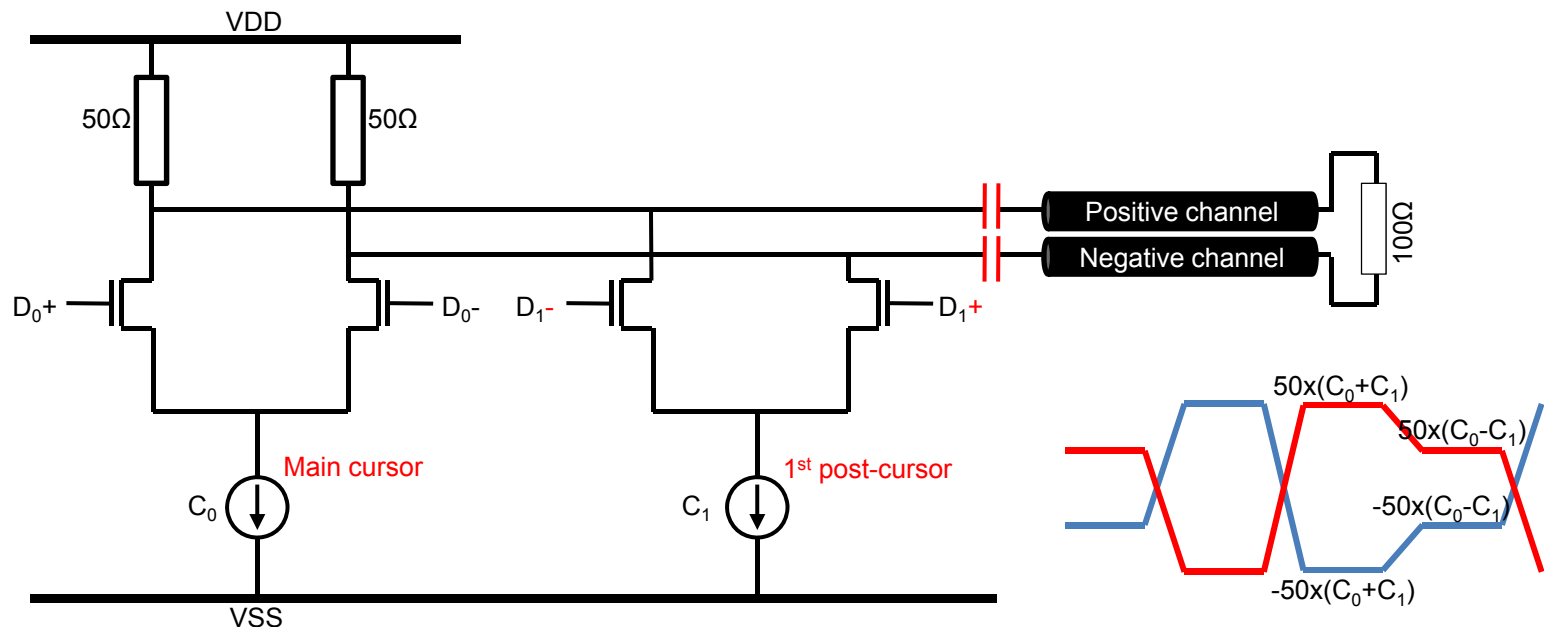
Circuit implementation

- Current-mode drivers can be easily used for pre-/de-emphasis
 - It is very easy to modify drivers into current-mode adder including controllable gain



Circuit implementation

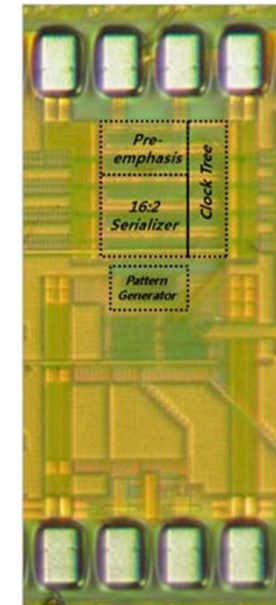
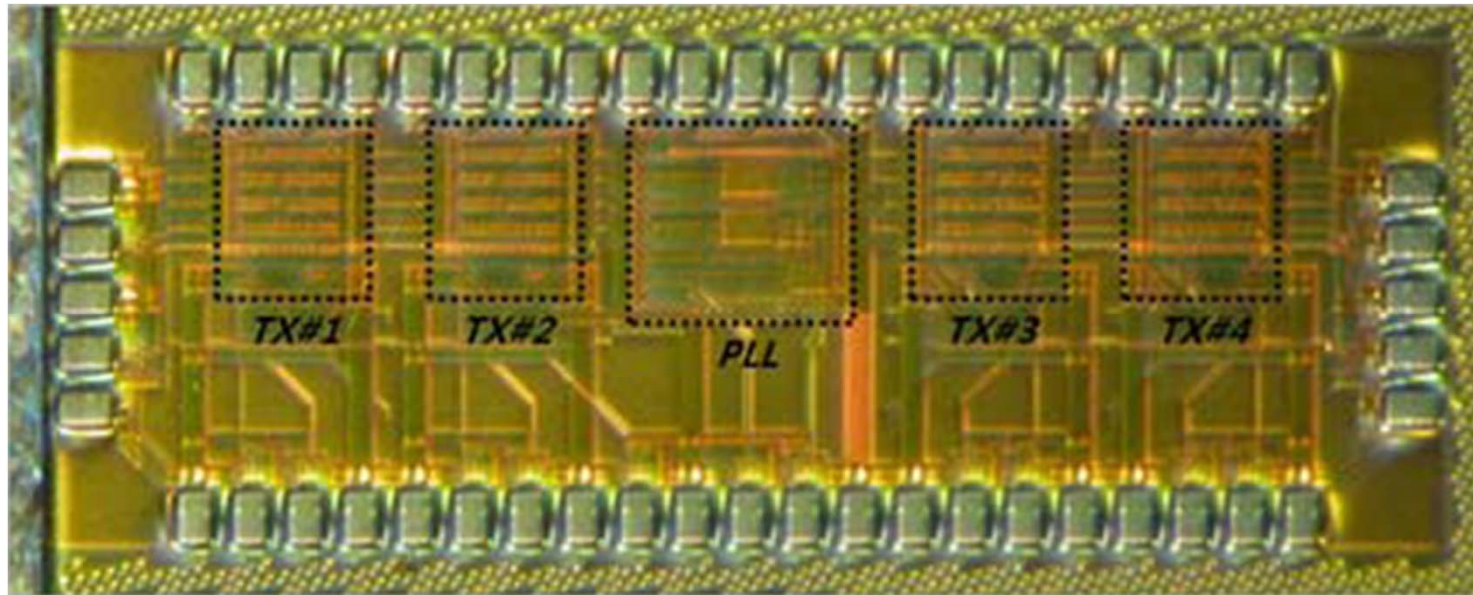
- Simultaneous implementation of pre-/de-emphasis
 - $D_1 = D_0 \rightarrow V_{\text{out,diff}} = \pm 50 \times (C_0 - C_1) \rightarrow \text{De-emphasis}$
 - $D_1 \neq D_0 \rightarrow V_{\text{out,diff}} = \pm 50 \times (C_0 + C_1) \rightarrow \text{Pre-emphasis}$
 - Level difference is defined as sum and subtract



Tx- vs. Rx- equalization

- Tx equalization
 - Consumes large power
 - Enlarged output signal improves SNR at Rx side
 - Easy implementation
- Rx equalization
 - Relatively low power consumption
 - More complex implementation (especially DFE)
 - For best performance, LE and DFE combination

Design example



“4-Channel 3.2/6.4-Gbps Dual-rate Transmitter”

김두호, 최우영

대한전자공학회 논문지 2010

4ch transmitter with 1-tap pre-emphasis

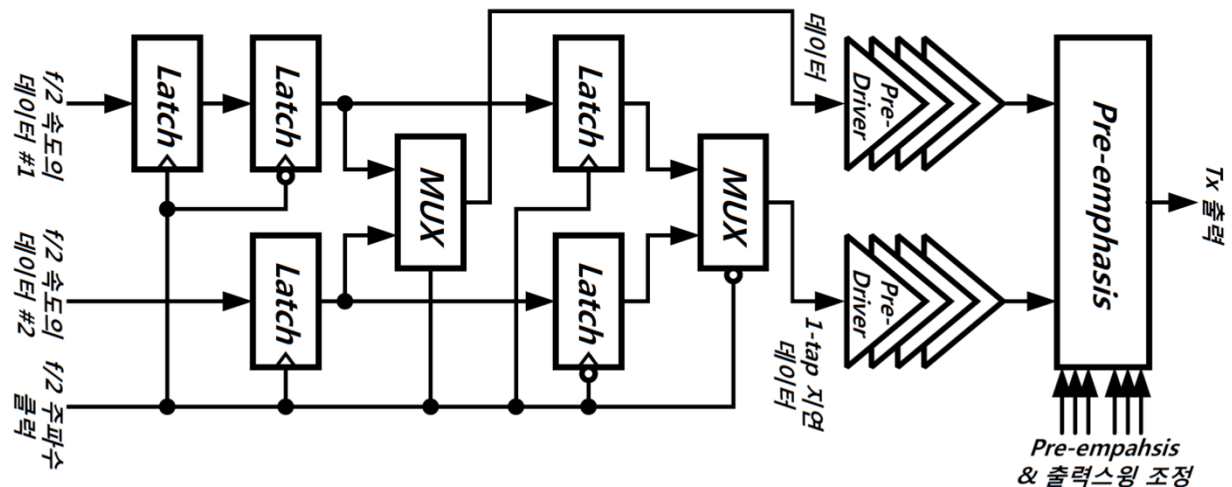
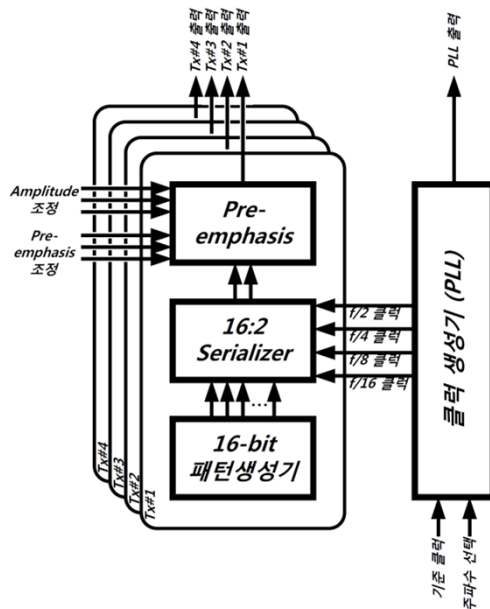
Dual-rate (3.2/ 6.4 Gbps)

130nm CMOS technology / COB package

600mW dissipation @1.2V power supply

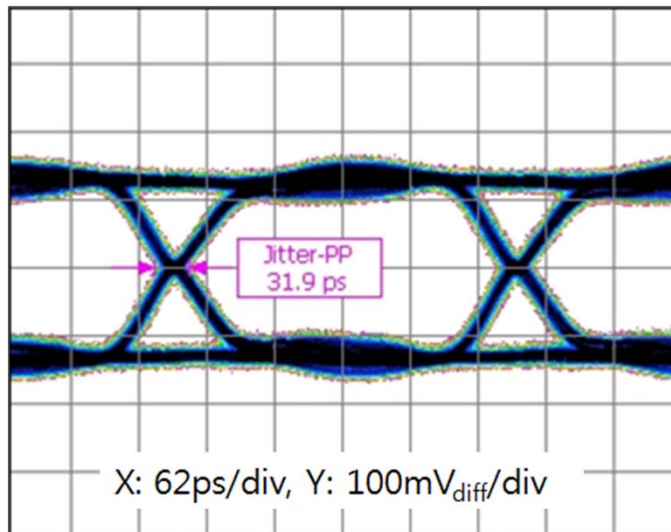
Design example

- 4-channel transmitter sharing a clock generator
 - 2:1 serializer function is included in pre-emphasis circuit
 - Displayport application

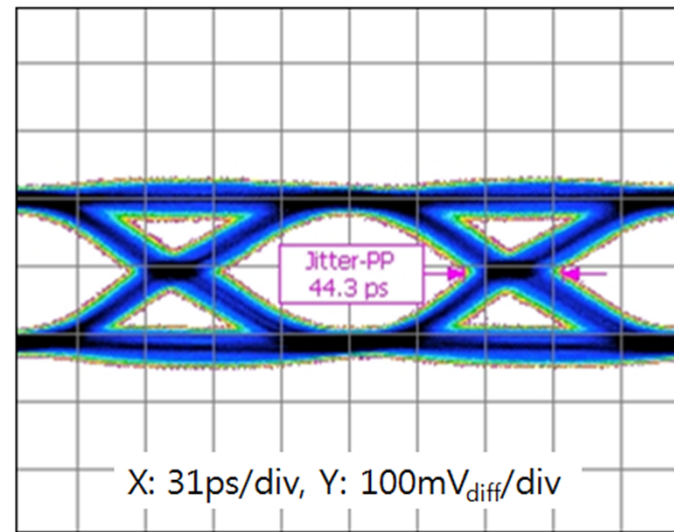


Design example

- Clock generator performance
 - PLL jitter is main performance metric of transmitter evaluation.



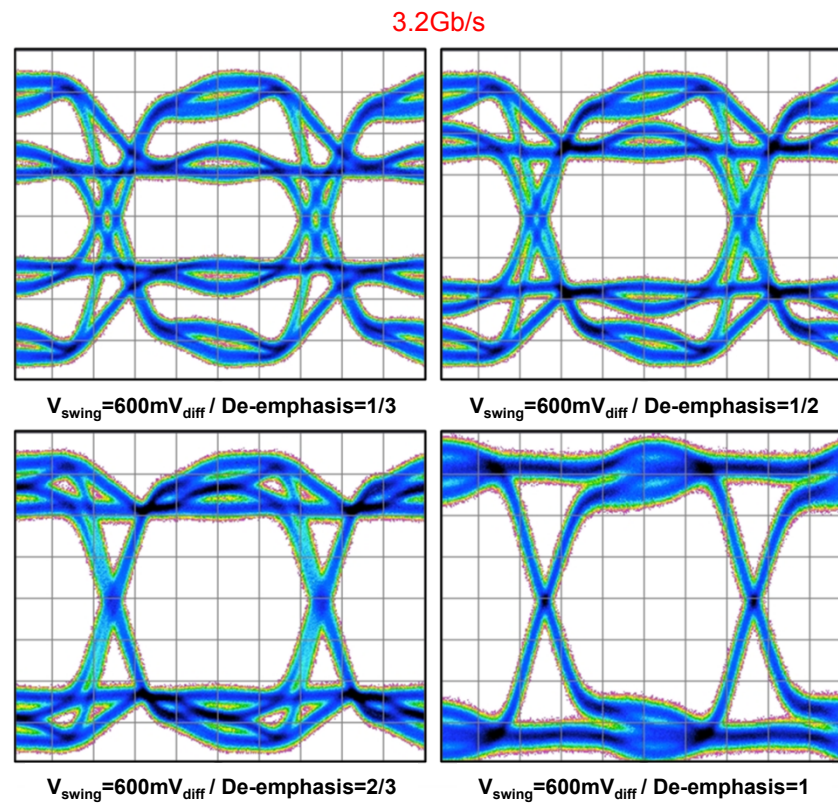
3.2Gb/s



6.4Gb/s

Design example

- De-emphasis waveform



Design example

- De-emphasis waveform

